

Water Supply and Resident Feedback Analysis

Multi-Floor Residential Facility • Accra, Ghana

Data collected via anonymous Google Forms survey, April 2026. No personally identifying information recorded.

129 residents surveyed • Prepared by Michael Arkorful

Executive Summary

A multi-floor residential facility in Accra faces a critical water supply failure: 89% of residents experience shortages daily, 80% endure outages lasting 3 hours or more, and the average negative impact score is 4.0 out of 5. Satisfaction with management communication has collapsed to 1.6 out of 5, with 88% of reported issues going unresolved or unreported entirely. The most actionable finding is that residents' primary frustration is unpredictability - 72% say outages follow no pattern, and the top suggestions are a consistently stable water supply and a fixed schedule.

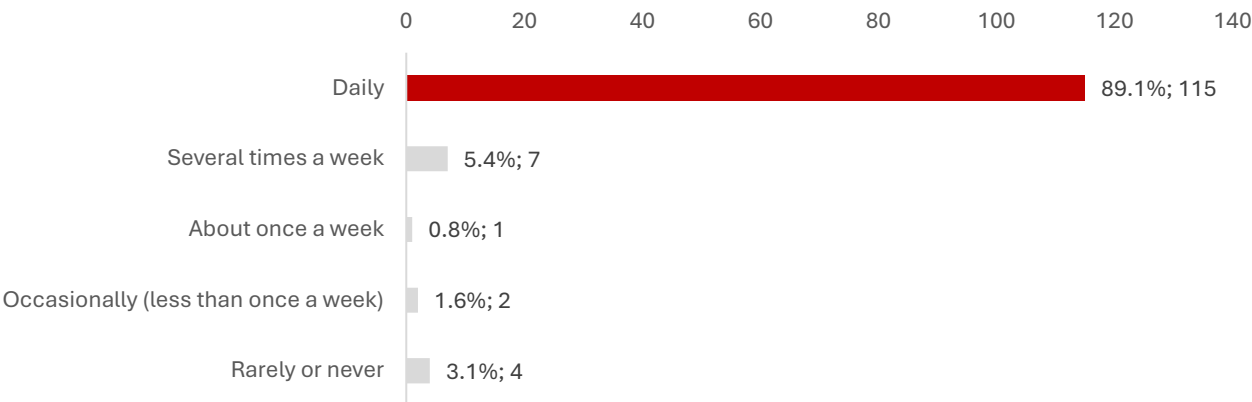
Outcome: All three recommendations from this analysis were implemented by building ownership: a zero-cost water schedule was published immediately, a new borehole was drilled, and a multi-stage filtration system was contracted for installation.

Key Findings

1. Shortage Frequency

89% (about 9 out of 10) of our residents indicated experiencing daily water outages. This is not a one-off situation but a persistent failure of a basic service.

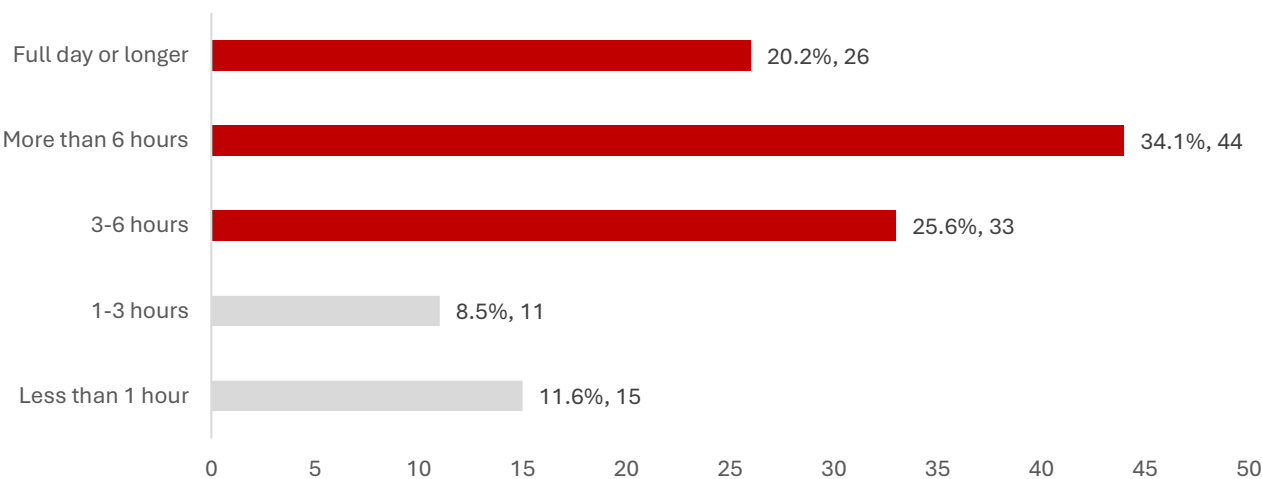
How often do residents experience water shortages?



2. Outage duration

Residents experience severe water outages, with 80% indicating a duration of 3 hours or more and 34.1% reporting an outage exceeding 6 hours. 20.2% experience a full day without water.

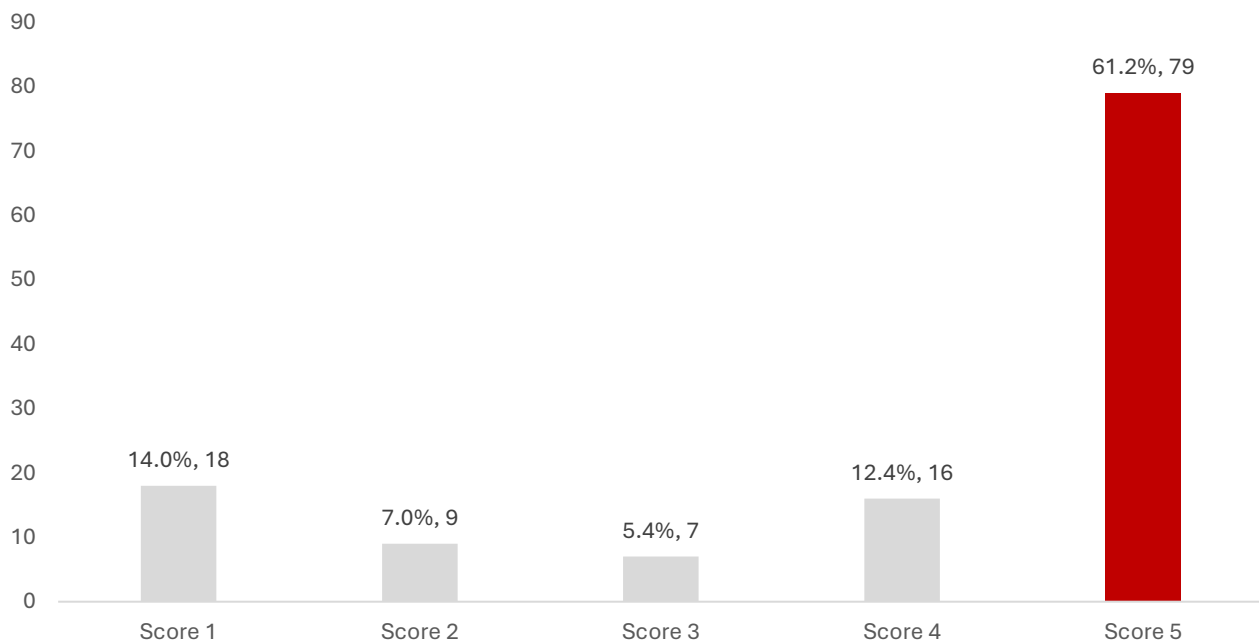
How long do water outages typically last?



3. Impact distribution

61% of residents gave the maximum negative impact score of 5 out of 5. The distribution is heavily skewed toward severe - this is not a divided opinion. The building has near-consensus that the problem is serious.

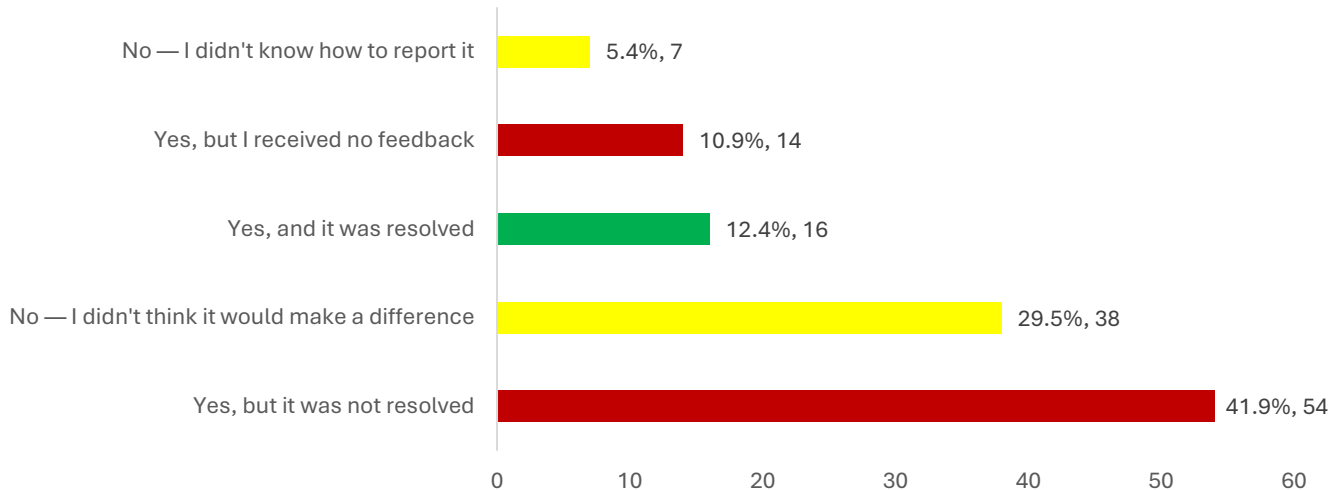
Overall negative impact on daily life (1 = low, 5 = severe)



4. Reporting experience

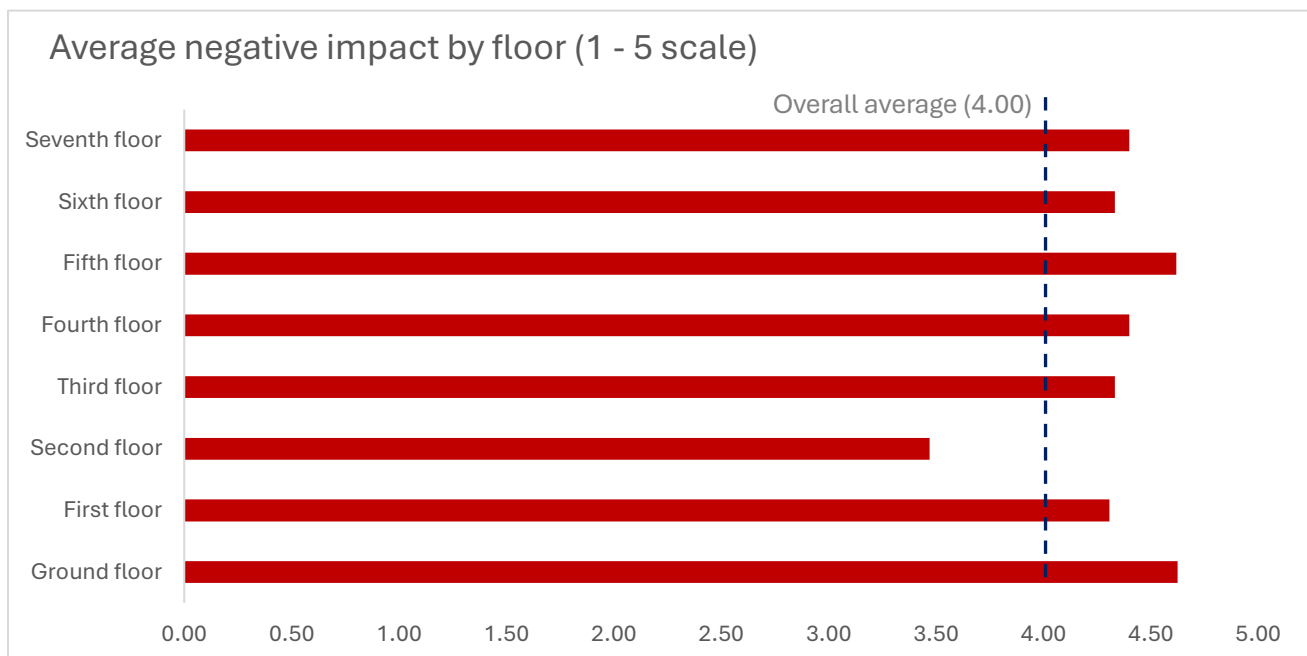
88% of residents who reported issues saw no resolution, received no feedback, or stopped reporting because they expected nothing would change. Only 12% experienced a successful resolution. The reporting system has effectively broken down.

What happens when residents report water issues?



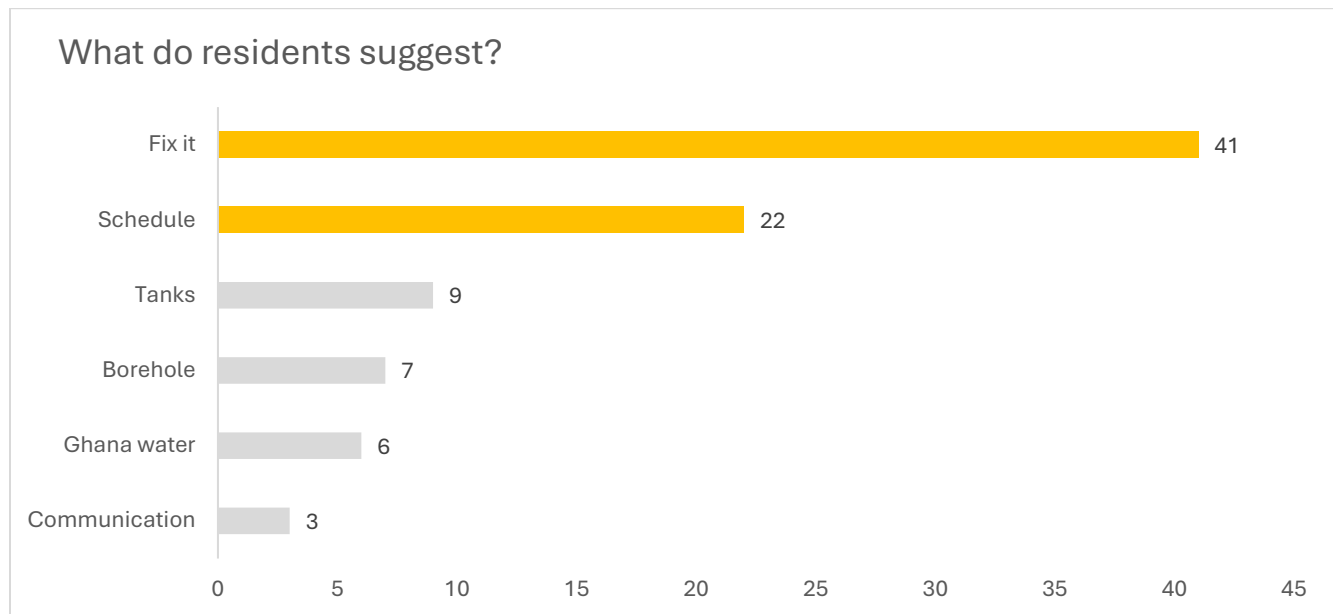
5. Floor-by-floor impact

Impact scores are uniformly high across all eight floors (with an average of 4.00). There is no floor-specific pattern. This rules out a pressure or distribution problem and points to a building-wide supply shortage.



6. Suggestion themes

Among residents who suggested a specific solution, a general fix of the problem (consistent water supply) and a water schedule were the most common request - more than twice as frequent as any infrastructure suggestion. Residents are asking for predictability and more water.



Recommendations

1. Immediate (zero cost): Implement a fixed water schedule and communicate it via WhatsApp group. 72% of residents say shortages are unpredictable, and 93% want WhatsApp or any communication channel. A published schedule directly addresses one of the highest complaints and costs nothing.
2. Invest in a second borehole or connect to the Ghana Water Company. The floor-by-floor analysis confirms this is a supply problem, not a distribution problem.
3. Install multi-stage filtration and purification systems to address water quality complaints - 5 residents reported brown or discolored water, causing skin reactions.

Outcome

This report was presented to building ownership and the facility management team. All three recommendations were implemented:

1. A fixed water schedule was published via WhatsApp group (immediate, zero cost)
2. A new borehole was drilled to increase water supply capacity
3. Experts were contracted to install a multi-stage filtration system

Secondary issues (appendix)

While this analysis focuses on water supply, residents raised additional concerns in open-text responses: gas schedule restrictions (9 mentions), WiFi not provided as promised (8 mentions), maintenance issues (9 mentions), staff concerns (5 mentions), elevator reliability (4 mentions). These are documented for management awareness.

Methodology

Data collected via an anonymous Google Forms survey distributed to residents of a multi-floor residential facility in Accra, Ghana. 129 responses collected over 4 days in April 2026. Survey included structured ratings (1–5 Likert scale), single and multi-select categorical questions, and 3 optional open-text fields. No personally identifying information was collected.

Analysis performed in Microsoft Excel using COUNTIF, COUNTIFS, AVERAGEIF, and wildcard matching for multi-select fields. Open-text responses were manually theme-coded to extract recurring patterns. Six visualizations were created following data-ink ratio and decluttering best practices.

Limitations: Self-selection bias is likely. Satisfaction scores are subjective perceptions. Theme coding was performed manually; counts are approximate.